CLOUD COMPUTING BASICS

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1. CLOUD COMPUTING BASICS

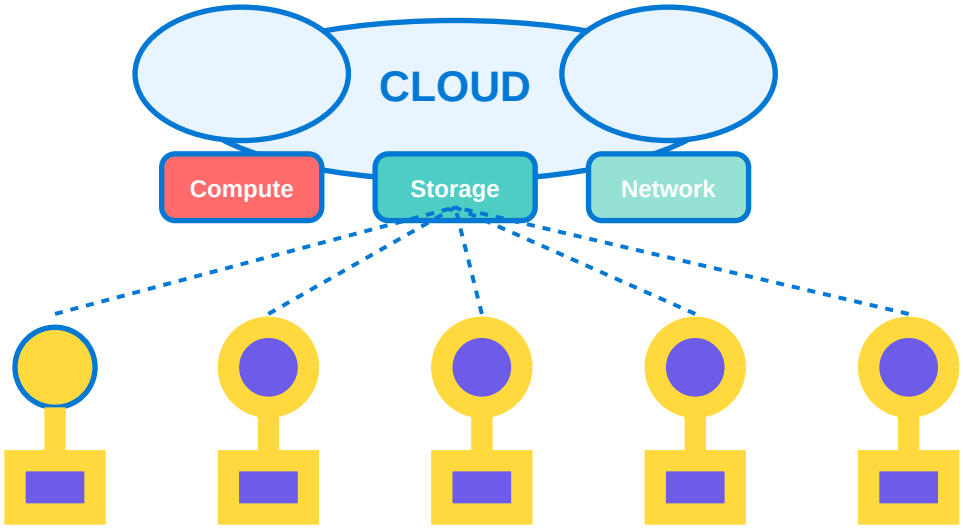
Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale.

Organizations typically pay only for cloud services they use, helping lower operating costs, run infrastructure more efficiently, and scale as business needs change. Cloud computing eliminates the capital expense of buying hardware and software and setting up and running on-site datacenters.

Key Characteristics:

On-Demand Self-Service	Users can provision computing capabilities automatically without human interac
Broad Network Access	Capabilities available over network and accessed through standard mechanism
Resource Pooling	Provider's resources are pooled to serve multiple consumers using multi-tenant
Rapid Elasticity	Capabilities can be elastically provisioned and released to scale rapidly
Measured Service	Cloud systems automatically control and optimize resource use

Cloud Computing Architecture



Users accessing cloud services from various devices

2. CLOUD SERVICE MODELS

Cloud computing offers different service models to meet various business requirements. The three primary service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS).

Infrastructure as a Service (IaaS):

IaaS provides virtualized computing resources over the internet. It offers maximum flexibility and management control over IT resources. Users can rent virtual machines, storage, and networks on a pay-as-you-go basis.

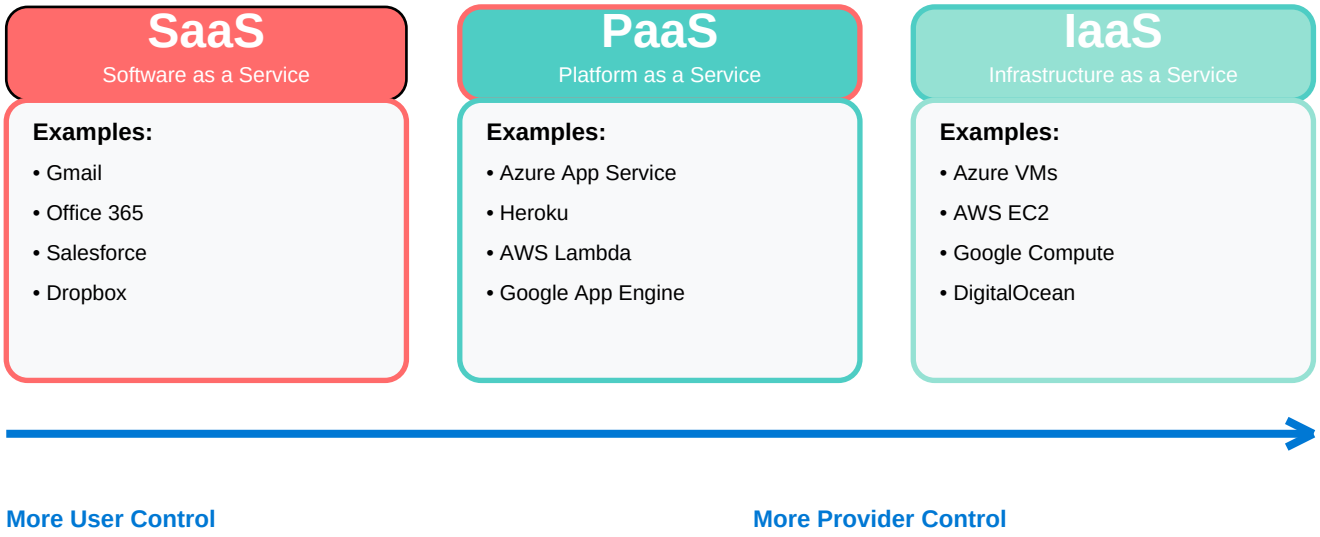
Platform as a Service (PaaS):

PaaS provides a platform allowing customers to develop, run, and manage applications without the complexity of building and maintaining the infrastructure. It includes development tools, database management systems, and business analytics.

Software as a Service (SaaS):

SaaS delivers software applications over the internet, on a subscription basis. The cloud provider hosts and manages the software application and underlying infrastructure, and handles maintenance, such as software upgrades and security patching.

Cloud Service Models (SaaS, PaaS, IaaS)



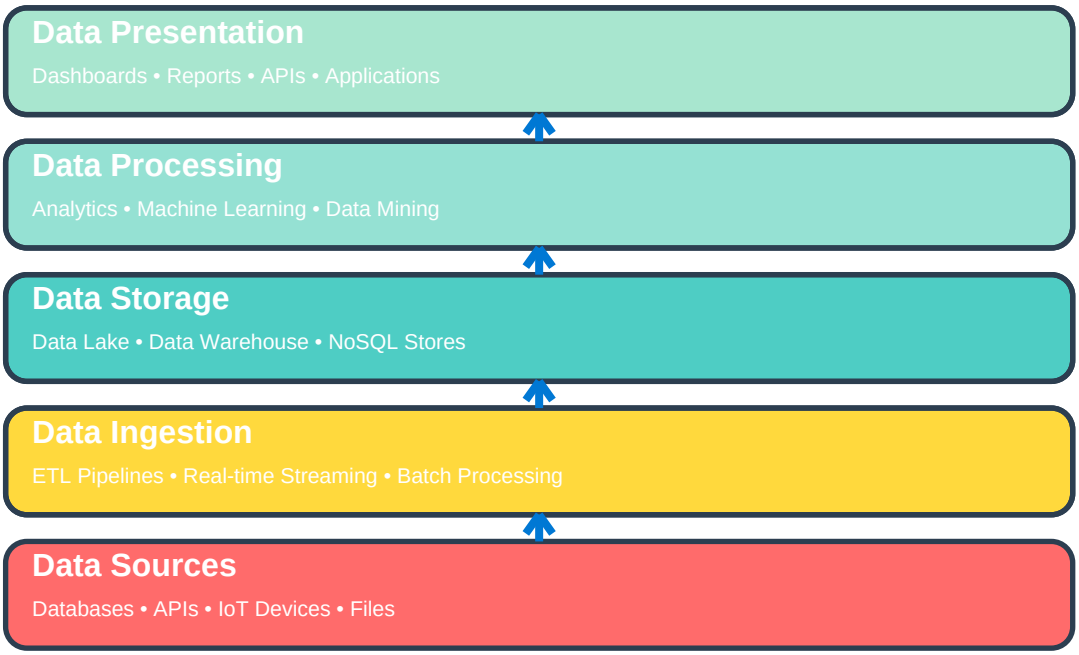
3. DATA ARCHITECTURE

Data architecture is a framework for how data is collected, stored, transformed, distributed, and consumed across an organization. It defines the blueprint for managing data assets and outlines how data flows through different systems and processes.

Components of Data Architecture:

Data Sources	Origin points of data including databases, APIs, IoT devices, and external systems
Data Ingestion	Process of importing, transferring, loading and processing data for later use
Data Storage	Repositories like data lakes, data warehouses, and databases for storing data
Data Processing	Transformation, analysis, and enrichment of data using various tools and techniques
Data Presentation	Visualization and consumption through dashboards, reports, and applications

A well-designed data architecture ensures data quality, accessibility, security, and compliance with regulations. It enables organizations to make data-driven decisions and supports advanced analytics and machine learning initiatives.



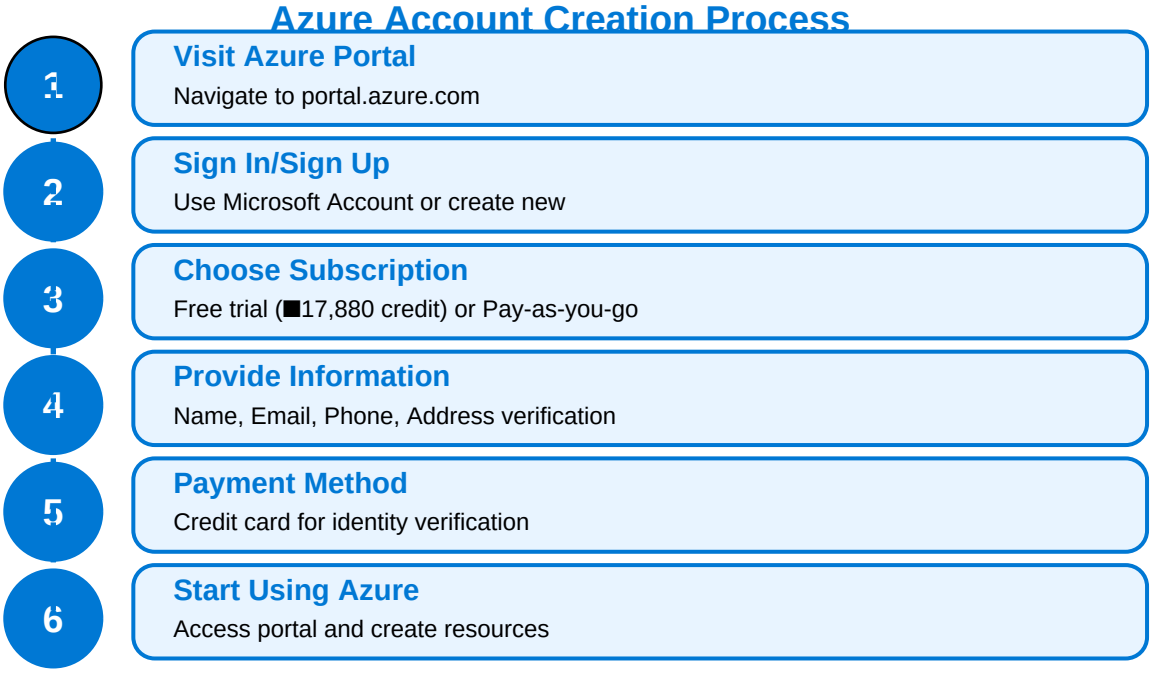
4. AZURE ACCOUNT CREATION

Microsoft Azure is a cloud computing platform that offers a wide range of cloud services including compute, analytics, storage, and networking. Creating an Azure account is the first step to accessing these services.

Prerequisites:

- A valid email address (Microsoft account or work/school account)
- A valid phone number for verification
- A credit or debit card for identity verification (not charged unless you upgrade)
- Valid address information

Account Creation Steps:



Free Trial Benefits: Azure offers a free trial with ■17,880.69 (or $200) in credit that can be used within 30 days. Additionally, you get 12 months of popular services for free and access to 25+ services that are always free.

Azure Portal Dashboard:

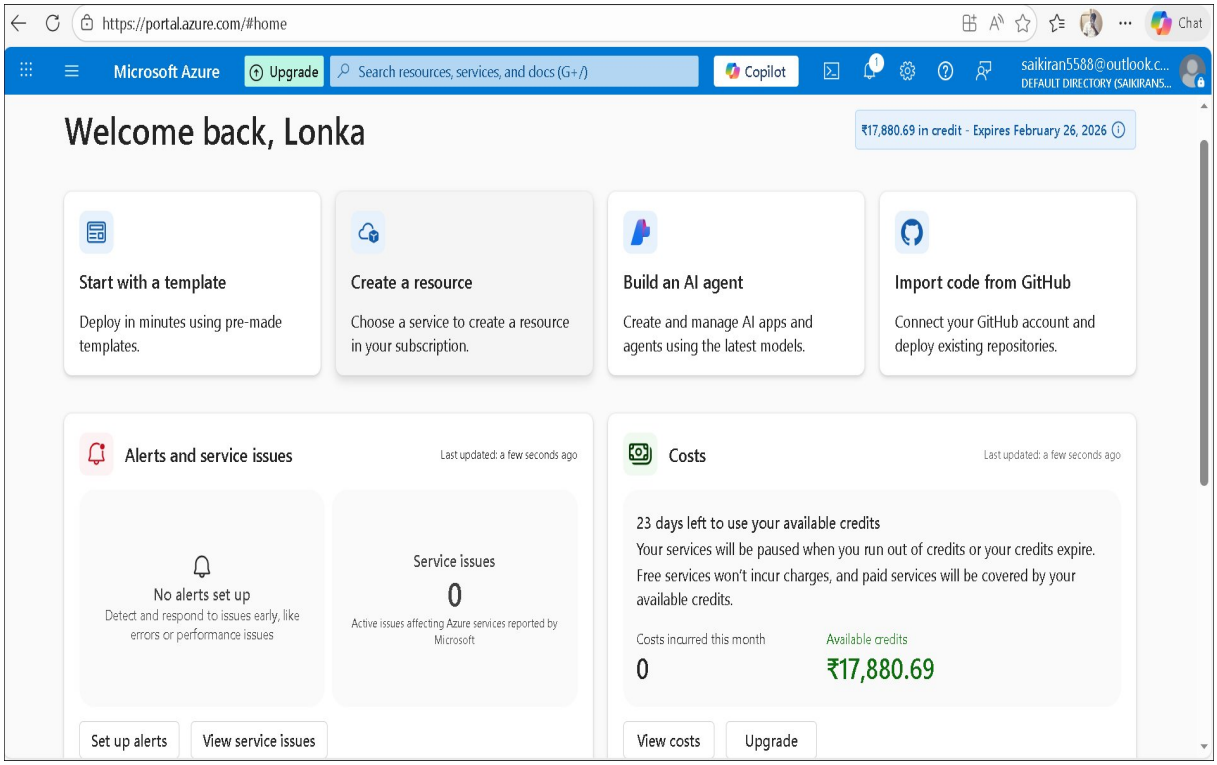


Figure: Azure Portal showing account dashboard with credit information

5. AZURE PORTAL OVERVIEW

The Azure Portal is a web-based, unified console that provides an alternative to command-line tools. With the Azure Portal, you can manage your Azure subscription using a graphical user interface.

Key Features of Azure Portal:

Dashboard	Customizable view of your Azure resources and services
Resource Management	Create, configure, and manage all your Azure resources
Cost Management	Monitor spending and manage budgets with cost analysis tools
Azure Copilot	AI-powered assistant to help navigate and use Azure services
Cloud Shell	Integrated command-line interface for managing resources
Monitoring & Alerts	Real-time monitoring of resources and setting up alert notifications

Common Portal Actions:

- **Create Resources:** Deploy virtual machines, databases, web apps, and other services
- **Configure Settings:** Manage security, networking, and access control
- **Monitor Performance:** View metrics, logs, and diagnostics
- **Manage Costs:** Track spending and optimize resource usage
- **Access Documentation:** Built-in help and learning resources

The portal interface is organized into several sections including Home, All services, Resource groups, Subscriptions, and Settings. Navigation is intuitive with a search bar at the top for quickly finding services and resources.

6. CONCLUSION

Cloud computing has revolutionized the way organizations manage and deliver IT services. By leveraging cloud platforms like Microsoft Azure, businesses can achieve greater agility, scalability, and cost-efficiency.

Understanding cloud service models (IaaS, PaaS, SaaS) enables organizations to choose the right level of control and management for their specific needs. A well-designed data architecture ensures that data flows efficiently through the organization, supporting analytics and decision-making processes.

Microsoft Azure provides a comprehensive suite of cloud services with an intuitive portal interface. The free trial option with substantial credits allows students and professionals to explore and learn Azure services without initial investment.

Key Takeaways:

1	Cloud computing offers flexibility, scalability, and cost savings
2	Different service models (IaaS, PaaS, SaaS) serve different business needs
3	Data architecture is fundamental for managing data lifecycle
4	Azure provides enterprise-grade cloud services with global presence
5	Azure Portal offers user-friendly interface for resource management
6	Free trial credits enable hands-on learning and experimentation

As cloud adoption continues to grow, proficiency in cloud platforms like Azure becomes increasingly valuable. This document has provided foundational knowledge in cloud computing concepts, data architecture principles, and practical steps for getting started with Microsoft Azure.

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